CREST CPSA TECHNICAL SPECIFICATIONS QUICK SHEET

Complete consolidated reference: Wireless, Ciphers, Hashes, Addressing & Subnetting

1. WIRELESS STANDARDS, KEYS & IV SPECIFICATIONS

Standard / Protocol	Key Size	IV / Nonce Size	Total Effective Key	Key Notes
WEP (Standard / 64-bit)	40 bits	24 bits	64 bits	Weak RC4 implementation; IV space too small (leads to IV collision attacks)
WEP (Extended / 128-bit)	104 bits	24 bits	128 bits	Uses identical 24-bit IV size; vulnerable to statistical key recovery
TKIP (WPA)	128 bits	48 bits (TSC)	176 bits	Stopgap protocol; 48-bit Sequence Counter eliminates IV reuse
CCMP (WPA2)	128 bits	48 bits (PN)	176 bits	AES-based; uses a 48-bit Packet Number as nonce

2. CIPHERS, BLOCK SIZES & KEY SPECIFICATIONS

Algorithm	Category	Key Size	Block / Cycle Details
DES	Symmetric Block Cipher	56 bits	64-bit block size; 16 cycles of 48-bit subkeys
3DES (Triple DES)	Symmetric Block Cipher	168 bits	3 × 56-bit keys; 48 cycles
AES	Symmetric Block Cipher	128, 192*, or 256 bits	128-bit block size (*PDF misprints 192 as 129)
RC4	Symmetric Stream Cipher	Variable (typically 40–256 bits)	Used in original WEP and legacy NTLM
Blowfish / Twofish / Serpent	Symmetric Block Cipher	Variable (up to 448 / 256 bits)	CPSA symmetric algorithm reference list
RSA / ElGamal / ECC	Asymmetric Encryption	Typically 1024–4096 bits (ECC: 256–521)	Public-key cryptosystems
Diffie-Hellman	Asymmetric Key Exchange	Parameter group size (e.g., DH 2048)	Key agreement protocol

3. CRYPTOGRAPHIC HASHES & AUTHENTICATION OUTPUT LENGTHS

Hash / Format	Output (Bytes)	Output (Bits)	Hex Length	Algorithm Details
MD5	16 bytes	128 bits	32 hex chars	Digest length; used in NTLMv2
SHA-1	20 bytes	160 bits	40 hex chars	Digest length
SHA-256	32 bytes	256 bits	64 hex chars	Digest length
SHA-512	64 bytes	512 bits	128 hex chars	Digest length
LM Hash (LANMAN)	16 bytes (total)	128 bits	32 hex chars	Max input limit: 14 chars; split into two 7-byte DES operations
NTLM Hash	16 bytes	128 bits	32 hex chars	Stored Windows password hash based on MD4/RC4
NTLMv2	16 bytes	128 bits	32 hex chars	HMAC-MD5 response format
MySQL < 4.1 Hash	8 bytes	64 bits	16 hex chars	Legacy custom password hash
MySQL 5+ Hash	20 bytes	160 bits	40 hex chars	SHA-1(SHA-1(password)), prefixed by '*'

4. NETWORK ADDRESSING, TTLS & VLAN BOUNDS

Protocol / Parameter	Bit Length	Byte Length	Format / Range	Notes
MAC Address	48 bits	6 bytes	12 hex digits	Physical layer 2 hardware address
IPv4 Address	32 bits	4 bytes	Dotted decimal (e.g. 192.168.0.1)	Network layer logical address
IPv6 Address	128 bits	16 bytes	32 hex digits (8 groups of 4)	No broadcast; uses anycast and multicast
Port Address Space	16 bits	2 bytes	0 to 65,535 (65,536 total)	Separate port spaces for TCP and UDP
TTL: Linux	8 bits	1 byte	Default value: 64	Diagnostic identifier for OS fingerprinting
TTL: Windows	8 bits	1 byte	Default value: 128	Diagnostic identifier for OS fingerprinting
TTL: Solaris / Network	8 bits	1 byte	Default value: 255	Diagnostic identifier for OS fingerprinting
Normal VLAN Range	12 bits	—	IDs 1 to 1005	Stored in vlan.dat
Legacy Media VLANs	—	—	IDs 1002 to 1005	Reserved for Token Ring and FDDI
Extended VLAN Range	12 bits	—	IDs > 1005 (up to 4094)	Not stored in vlan.dat

5. CLASSFUL IP SUBNETTING SIZING & BIT MASKS

Class	Leading Bits	Prefix	Subnet Mask	Usable Networks	Usable Hosts per Network
Class A	0 (1st bit 0)	/8	255.0.0.0	$2^7 = 128$	$2^{24} - 2 = 16{,}777{,}214$
Class B	10 (2nd bit 0)	/16	255.255.0.0	$2^{14} = 16{,}384$	$2^{16} - 2 = 65{,}534$
Class C	110 (3rd bit 0)	/24	255.255.255.0	$2^{21} = 2{,}097{,}152$	$2^8 - 2 = 254$
Class D	1110 (4th bit 0)	/4	N/A (Multicast)	N/A	Reserved for Multicast
Class E	1111 (4th bit 1)	/4	N/A (Experimental)	N/A	Experimental / Research space